

Shamsa Abid, Ph.D.

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 <https://shamsaabid.github.io/>

Research Interests

- AI-powered software engineering; program analysis and reasoning with large language models; trustworthiness and explainability of code generation systems; code recommendation and reuse; mining software repositories; empirical SE; design patterns and algorithms.

Postdoc Employment

- Oct 2024 – Aug 2025  **Assistant Professor** National University of Computer and Emerging Sciences (NUCES)
- May 2022 – May 2024  **Research Scientist** Singapore Management University, *Singapore*
- Dec 2021 – Mar 2022  **Research Intern** JetBrains, *Czech Republic*

Education

- 2015 – 2021  **Ph.D. Computer Science, LUMS**
Thesis title: *Feature-driven API Usage based Code Recommendation for Opportunistic Reuse*
- 2011 – 2013  **MS Computer Science, LUMS**
Project title: *Low Literacy Keyboard for Mobile Devices: Enabling Fast and Efficient Text Input.*
- 2000 – 2004  **BS Hons. Computer Science, Lahore College for Women University**
Project title: *Library Information Management System.*

Teaching Experience

- Spring 2025  **Assistant Professor** *Software Reengineering* , NUCES
-  **Assistant Professor** *Design and Analysis of Algorithms* , NUCES
- Fall 2024  **Assistant Professor** *Software Requirements Engineering* , NUCES
-  **Assistant Professor** *Data Structures* , NUCES
- Spring 2021  **Teaching Assistant** *Software Development Tools and Processes* , LUMS
- Spring 2020  **Teaching Assistant** *Software Development Tools and Processes* , LUMS
- Fall 2018  **Visiting Assistant Professor** *Web Programming* , NUCES
- Spring 2015  **Visiting Lecturer** *Software Quality Assurance* , Kinnaird College for Women University
-  **Teaching Assistant** *Software Engineering* , LUMS
- Fall 2014  **Teaching Assistant** *Design Patterns and Refactoring* , LUMS
- Fall 2013  **Teaching Assistant** *Design Patterns and Refactoring* , LUMS
- Spring 2013  **Teaching Assistant** *Software Engineering* , LUMS
- Fall 2012  **Visiting Lecturer** *Creating Digital Content* , Beaconhouse National University
-  **Teaching Assistant** *Introduction to Computing* , LUMS

Industry Experience

- Aug 13 – Oct 14  **Senior Software Engineer** Techlogix Pvt. Ltd.
Apr 07 – Sep 07  **Software Engineer** Xavor Pvt. Ltd.

Research Experience

- Jan 13 – Jun 13  **Research Associate** *Software Testing* LUMS
Jun 11 – Aug 11  **Research Associate** *Technology for Low Literacy Segment*, LUMS

Research Publications

Journal Articles

- 1 S. Abid, X. Cai, and L. Jiang, “Measuring model alignment for code clone detection using causal interpretation,” *Empirical Software Engineering*, vol. 30, no. 2, p. 46, 2025.
- 2 S. Abid, S. Shamail, H. A. Basit, and S. Nadi, “FACER: An API usage-based code-example recommender for opportunistic reuse,” *Empirical Software Engineering*, vol. 26, no. 6, p. 110, 2021.

Conferences and Workshops

- 1 S. Kaneez Fatima, Y. Abrar, A. Rehman, A. Nawaz, S. Abid, and A. A. Bangash, “On the Footprints of Reviewer Bots’ Feedback on Agentic Pull Requests in OSS GitHub Repositories,” in *2026 IEEE 23rd International Mining Software Repositories Conference (MSR)*, 2026.
- 2 M. T. Jamil, S. Abid, and S. Shamail, “Can LLMs Generate Higher Quality Code Than Humans? An Empirical Study,” in *2025 IEEE/ACM 22nd International Conference on Mining Software Repositories (MSR)*, 2025, pp. 478–489.  DOI: 10.1109/MSR66628.2025.00081.
- 3 S. Abid, X. Cai, and L. Jiang, “Interpreting CodeBERT for Semantic Code Clone Detection,” in *2023 30th Asia-Pacific Software Engineering Conference (APSEC)*, IEEE, 2023, pp. 229–238.
- 4 S. Abid, H. Abdul Basit, and S. Shamail, “Context-aware code recommendation in IntelliJ IDEA,” in *Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, 2022, pp. 1647–1651.
- 5 S. Arshad, S. Abid, and S. Shamail, “CodeBERT for code clone detection: A replication study,” in *2022 IEEE 16th International Workshop on Software Clones (IWSC)*, IEEE, 2022, pp. 39–45.
- 6 S. M. Rabbani, N. A. Gulzar, S. Arshad, S. Abid, and S. Shamail, “A comparative analysis of clone detection techniques on SemanticCloneBench,” in *2022 IEEE 16th International Workshop on Software Clones (IWSC)*, IEEE, 2022, pp. 16–22.
- 7 M. Kamal, A. Abaid, S. Abid, and S. Shamail, “FACER-AS: An API Usage-based Code Recommendation Tool for Android Studio,” in *2021 IEEE International Conference on Software Maintenance and Evolution (ICSME)*, IEEE, 2021, pp. 609–613.
- 8 S. Abid, “Recommending related functions from API usage-based function clone structures,” in *Proceedings of the 2019 27th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, ser. ESEC/FSE 2019, Tallinn, Estonia: Association for Computing Machinery, 2019, pp. 1193–1195, ISBN: 9781450355728.  DOI: 10.1145/3338906.3342486.
- 9 S. Abid, S. Javed, M. Naseem, S. Shahid, H. A. Basit, and Y. Higo, “CodeEase: Harnessing method clone structures for reuse,” in *2017 IEEE 11th International Workshop on Software Clones (IWSC)*, IEEE, 2017, pp. 1–7.

- 10 S. Abid and H. A. Basit, "Towards a structural clone based recommender system," in *2016 IEEE 23rd International Conference on Software Analysis, Evolution, and Reengineering (SANER)*, IEEE, vol. 3, 2016, pp. 51–52.
- 11 S. Abid, Z. Qamar, N. Khan, M. Shayan, and H. A. Basit, "Retrieving Design Pattern Usage Examples using Domain Matching," in *2016 23rd Asia-Pacific Software Engineering Conference (APSEC)*, IEEE, 2016, pp. 33–40.
- 12 S. Abid, H. Abdul Basit, and N. Arshad, "Reflections on Teaching Refactoring: A Tale of Two Projects," in *Proceedings of the 2015 ACM Conference on Innovation and Technology in Computer Science Education*, ser. ITiCSE '15, Vilnius, Lithuania: Association for Computing Machinery, 2015, pp. 225–230, ISBN: 9781450334402.  DOI: 10.1145/2729094.2742617.

Seminars

- | | | |
|----------|---|--|
| Nov 2023 |  | <i>Why P not Q? Interpreting Code Models Decision Making</i> , LUMS |
| Oct 2023 |  | <i>Unveiling the Code Model Enigma: Interpreting the Gap Between AI and Human Intuition in Code Clone Detection</i> , Singapore Management University |
| Sep 2021 |  | <i>Comparing Clustering against Deep-Learning for Semantic Code Clone Detection</i> , LUMS |
| |  | <i>An API Usage-based Code Recommendation Tool for Android Studio</i> , LUMS |

Service

Program Committee Membership

- | | | |
|------|---|---|
| 2026 |  | Research Track <i>41st IEEE/ACM International Conference on Automated Software Engineering, ASE 2026</i> |
| 2025 |  | Research Track <i>40th IEEE/ACM International Conference on Automated Software Engineering, ASE 2025</i> |
| |  | Demonstrations Track <i>ACM International Conference on the Foundations of Software Engineering (FSE) 2025</i> |
| |  | Research Track <i>International Conference on Software Maintenance and Engineering (IC-SME) 2025</i> |
| 2024 |  | Research Track <i>International Conference on Software Maintenance and Engineering (IC-SME)</i> |
| |  | Technical Track <i>Mining Software Repositories (MSR)</i> |
| |  | Research Track <i>AI Foundation Models and Software Engineering (Forge)</i> |
| 2023 |  | Demonstrations Track <i>International Conference on Software Engineering (ICSE)</i> |
| |  | Industry Track <i>Mining Software Repositories (MSR)</i> |
| |  | Research Track <i>International Workshop on Interpretability and Robustness in Neural Software Engineering (InteNSE)</i> |
| 2021 |  | Shadow PC <i>Mining Software Repositories (MSR)</i> |

Journal Reviews

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|-------------|---|---|
| 2024 |  | <i>ACM Transactions on Software Engineering and Methodology (TOSEM)</i> |
| |  | <i>Empirical Software Engineering (EMSE)</i> |
| 2020 – 2021 |  | <i>Journal of Behavior and Information Technology</i> |

Awards and Certificates

Awards

- 2025  **Distinguished Reviewer Award**, Demonstrations Track, ACM, FSE 2025
- 2016  **Runner Up Poster**, SBASSE PhD Colloquium, LUMS
- 2013  **Deans Honor List Award**, LUMS
-  **Outstanding Teaching Assistant Award**, Software Engineering, LUMS

Certification

- 2026  **Agentic AI**. DeepLearning.AI.
- 2016  **Machine Learning Foundations: A Case Study Approach**. Coursera, University of Washington.
- 2012  **Intelligent Computing**, Course Audit Certificate, LUMS.
-  **iOS Development**, Workshop Participation Certificate, LUMS.
- 2005  **FOSSAC 2005, 1st National Free and Open Source Software Awareness Campaign Training**, FOSSFP: Free and Open Source Foundation Pakistan (FOSS Academy).
- 2004  **Internet and Multimedia Professional**, BrainNET.

References

Prof. Dr. Shafay Shamail, *Professor*, LUMS, sshamail@lums.edu.pk

Prof. Dr. Lingxiao Jiang, *Professor*, Singapore Management University, lxjiang@smu.edu.sg

Dr. Sarah Nadi, *Associate Professor*, New York University, Abu Dhabi, sarah.nadi@nyu.edu